

Group Member Contribution Declaration

I hereby declare that I actively participated in completing this group assignment and did not rely solely on the work of others. I understand that meaningful and honest contribution is essential for both the integrity of this assignment and my ability to successfully answer the mid-semester examination questions based on the DS Job Tracker survey work.

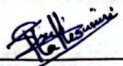
My specific contributions to this group assignment are as follows:

I contributed to finalizing the standardized set of variables for the final combined dataset. I cleaned the group 03 dataset to match this standardized structure, and I documented all the steps I followed during the cleaning and preparation process.

I certify that the above statements are true and reflect my genuine involvement in the task.

Name of Student: Y.G.D.M.S. Thathsarani

Index number: AS2022397

Signature: 

Date: 2025/11/20

I hereby declare that I actively participated in completing this group assignment and did not rely solely on the work of others. I understand that meaningful and honest contribution is essential for both the integrity of this assignment and my ability to successfully answer the mid-semester examination questions based on the DS Job Tracker survey work.

My specific contributions to this group assignment are as follows:

I contributed to finalizing the standardized set of variables for the final combined dataset. I cleaned the group 02 dataset to match this standardized structure, and I documented all the steps I followed during the cleaning and preparation process.

I certify that the above statements are true and reflect my genuine involvement in the task.

Name of Student: S.G.V.S. Samaranayaka

Index number: AS2022399

Signature: 

Date: 2025/11/20

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My specific contributions to this group assignment are as follows:

I contributed to finalizing the standardized set of variables for the final combined dataset. I cleaned the group 08 dataset to match this standardized structure, and I documented all the steps I followed during the cleaning and preparation process.

I certify that the above statements are true and reflect my genuine involvement in the task.

Name of Student: I.V.N.N. Wijesekara

Index number: AS2022404

Signature: N. V. N. N. Wijesekara

Date: 2025/11/20

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My specific contributions to this group assignment are as follows:

I contributed to finalizing the standardized set of variables for the final combined dataset. I cleaned the group 06 dataset to match this standardized structure, and I documented all the steps I followed during the cleaning and preparation process.

I certify that the above statements are true and reflect my genuine involvement in the task.

Name of Student: L.P.U.D.N. Vasana

Index number: AS2022407

Signature: L. P. U. D. N. Vasana

Date: 2025/11/20

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My specific contributions to this group assignment are as follows:

I contributed to finalizing the standardized set of variables for the final combined dataset. I cleaned the group 02 dataset to match this standardized structure, and I documented all the steps I followed during the cleaning and preparation process. Addition to that I created the data descriptions for the variables and helped finalize the combined dataset.

I certify that the above statements are true and reflect my genuine involvement in the task.

Name of Student: G.R. Jayasinghe

Index number: AS2022409

Signature: 

Date: 2025/11/20

I hereby declare that I actively participated in completing this group assignment and did not rely solely on the work of others. I understand that meaningful and honest contribution is essential for both the integrity of this assignment and my ability to successfully answer the mid-semester examination questions based on the DS Job Tracker survey work.

My specific contributions to this group assignment are as follows:

I contributed to finalizing the standardized set of variables for the final combined dataset. I cleaned the group 07 dataset to match this standardized structure, and I documented all the steps I followed during the cleaning and preparation process.

I certify that the above statements are true and reflect my genuine involvement in the task.

Name of Student: G.K.L.L.S. Premathilaka

Index number: AS2022426

Signature: 

Date: 2025/11/20

I hereby declare that I actively participated in completing this group assignment and did not rely solely on the work of others. I understand that meaningful and honest contribution is essential for both the integrity of this assignment and my ability to successfully answer the mid-semester examination questions based on the DS Job Tracker survey work.

My specific contributions to this group assignment are as follows:

I contributed to finalizing the standardized set of variables for the final combined dataset. I cleaned the group 06 dataset to match this standardized structure, and I documented all the steps I followed during the cleaning and preparation process.

I certify that the above statements are true and reflect my genuine involvement in the task.

Name of Student: P.A.H.N. Dharmasena

Index number: AS2022430

Signature: P.A.H.N. Dharmasena

Date: 2025/11/20

I hereby declare that I actively participated in completing this group assignment and did not rely solely on the work of others. I understand that meaningful and honest contribution is essential for both the integrity of this assignment and my ability to successfully answer the mid-semester examination questions based on the DS Job Tracker survey work.

My specific contributions to this group assignment are as follows:

I contributed to finalizing the standardized set of variables for the final combined dataset. I cleaned the group 05 dataset to match this standardized structure, and I documented all the steps I followed during the cleaning and preparation process. Addition to that I helped in finalizing the documentation work.

I certify that the above statements are true and reflect my genuine involvement in the task.

Name of Student: N.L.N. Kumarage

Index number: AS2022469

Signature: N.L.N. Kumarage

Date: 2025/11/20

I hereby declare that I actively participated in completing this group assignment and did not rely solely on the work of others. I understand that meaningful and honest contribution is essential for both the integrity of this assignment and my ability to successfully answer the mid-semester examination questions based on the DS Job Tracker survey work.

My specific contributions to this group assignment are as follows:

I contributed to finalizing the standardized set of variables for the final combined dataset. I cleaned the group 09 dataset to match this standardized structure, and I documented all the steps I followed during the cleaning and preparation process.

I certify that the above statements are true and reflect my genuine involvement in the task.

Name of Student: M.N.D. Gunarathna

Index number: AS2022556

Signature: 

Date: 2025/11/20

I hereby declare that I actively participated in completing this group assignment and did not rely solely on the work of others. I understand that meaningful and honest contribution is essential for both the integrity of this assignment and my ability to successfully answer the mid-semester examination questions based on the DS Job Tracker survey work.

My specific contributions to this group assignment are as follows:

I contributed to finalizing the standardized set of variables for the final combined dataset. I cleaned the group 10 dataset to match this standardized structure, and I documented all the steps I followed during the cleaning and preparation process. Addition to that I helped in finalizing the documentation work.

I certify that the above statements are true and reflect my genuine involvement in the task.

Name of Student: N.G.S. Minimuthu

Index number: AS2022577

Signature: 

Date: 2025/11/20

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My specific contributions to this group assignment are as follows:

I contributed to finalizing the standardized set of variables for the final combined dataset. I cleaned the group 04 dataset to match this standardized structure, and I documented all the steps I followed during the cleaning and preparation process. Addition to that I helped in finalizing the documentation work.

I certify that the above statements are true and reflect my genuine involvement in the task.

Name of Student: H.L.D.H. Sandamini

Index number: AS2022636

Signature: Arunisanda .

Date: 2025/11/20

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My specific contributions to this group assignment are as follows:

I contributed to finalizing the standardized set of variables for the final combined dataset. I cleaned the group 1 dataset to match this standardized structure, and I documented all the steps I followed during the cleaning and preparation process. Addition to that I helped finalize the combined dataset.

I certify that the above statements are true and reflect my genuine involvement in the task.

Name of Student: Y.M.A.P. Rajapaksha

Index number: AS2022642

Signature: Amali

Date: 2025/11/20

Data Cleaning Steps

The variables in Group 1 dataset were considered as the initial variables. It contained 103 variables. As the first step, the remaining nine datasets were divided among group members to identify any new variables which are not in the Group 1 dataset. Based on this, a new excel sheet was created including all the variables in Group 1 dataset along with identified new variables. Using this list of variables, ten separate Excel sheets, one for each original group dataset were then prepared to ensure consistent structure across all datasets.

The variables identified through this process served as a preliminary set. The final list of variables was determined only after all ten datasets were combined and examined together. The steps carried out by each student for their assigned dataset are summarized below, including tasks such as handling missing values, transforming data and reporting renamed column names.

Group 1

- Since we selected this dataset as the initial dataset, most of the selected variables were present.
- However, information related to these variables did not appear.
- Acturial and risk analysis, Adaptability, AGE LIMIT, Airflow, Analytical_Skills, AngularJS, ANOVA, Apache kafka, Apache_Beam, Apache, API, Argo, Automation, Azure, Biopharmaceutical development and manufacturing, biostatistics, Canva, Cassandra, CDISC ADaM data standard, CI/CD tools, Clinical research, Cloud computing, Cloud Data Service, Clustering, CMC, Creating Dashboards, CRM systems(salesforce/hubspot/zoho), CS/SE Fundamentals, D3, Dagster, Data Cleaning, Data Engineering Principles, Data Infrastructure, Data migration, data Security, Data structures, Data validation, Data_governance, Data_modeling, DataBriks, DAX, Decision Making, Design Data, DevOps, DigitalOcean, Docker, Documentation, DOE, EHR, Employment Type(PART TIME/FULL TME/internship/contract), Epidemiology, ERP, ETL/ELT, Experience scaling casual inference models, Exploratory Data Analysis (EDA), Familiar with dbt, Fast API, Fine-tuning, Flink, Forecasting, Foresee, Full-Stack Skills, GA4, GCP, GIS, Git, Github, GitLab, Global Exposure, Glue, Google Data Studio, Google Tag Manager, Healthcare Analytics, HTML, Hypothesis Testing, Innovalon, Interpretation skills, Jenkins, JMP, Julia, Jupyter notebooks,

keras, KNN, Kubeflow, Kubernetes, Langchain, Languages, Large Scale data, Leadership, Linear Regression, logical Thinking, Looker, LLM, M Queries, MAR, MariaDB, Matplotlib, Medallion_Lakehouse_Architecture, Meta data and master data management, Microsoft Fabric ecosystem, MicroStrategy, Mixed Models, MLflow, MLOps tools, MS Fabric, MS SQL Sever, MS_Office, MS_Outlook, MS_Teams, Multivariate analysis, Navigate ambiguity, Network performance monitoring, NLP, NodeJS, Numpy, ONE DRIVE, OOP(OBJECTED-ORIENTED PROGRAMMING), OpinionLab, Organizational skills, OTHERS(LANGUAGES), Out look, pandas, PL/SQL, Policy-level data interpretation, potofolio management, Power Apps, Power Pivot, Power_Query, Predictive & Modeling Techniques, Preferred_Notice, Presto, Project management, Purview, PyTorch, Qualtrics, R Markdown, Random Forest, Redis, Report Preparation, Retrieval-Augmented Generation (RAG), Sampling methods, SAP, Scikit, Scripy, Scrum, Seaborn, Seldon, Semantic Kernel, Share point, Shell scripting, sklearn, Snowflake, Solid grasp of applied statistics, Solr, Spark SQL, SPC Tools/Software, Springboot, SQL Server, SSIS, StakeholderManagementSkills, Stata, Stateless API, Statistical analysis, statistical tests, Streamlit, Synapse, Tecton, Template Management, Teradata, Terraform, Time Series Analysis, TrinetX, T-SQL, Variance Component analysis, VBA, Warehouse Architecture, Weight & Biases, Windows, Writing_Skills, XGBoost, zoom, Seniority Level, special in Bsc

- There was a column called *knowledge in*. This column contained skills that were not included as separate variables in the dataset. Therefore, we used R code to extract the skills appearing in this *knowledge in* column.
- Then, using R, we created new columns for the extracted skills, marking them as **1** if the skill appeared and **NA** otherwise.
- These are the R codes we used. [Script.R](#)
- Then, it was pasted into our prepared Excel format.
- In this dataset, the minimum salary, maximum salary, and the currency were in the same column as salary.
- Therefore, we separated it into three columns: Min_Salary, Max_Salary, and Currency.
- Since there were only 32 data points in the salary column, we performed this separation manually in an Excel worksheet.

Group 2

This document summarizes all cleaning and standardization steps applied to the dataset group 2 before merging with the remaining datasets.

1. Column Modifications

a) Column Merged: Team Handling → Leadership

The original dataset contained a column named Team Handling. Since this information reflects leadership responsibilities, it was merged into the Leadership column for consistency. Avoid duplication of concepts referring to managerial capability.

b) Column Merged: Data Handling & Reporting → Report Preparation

The dataset included a Data Handling and Reporting column. This content aligned closely with the purpose of the Report Preparation variable. Information was therefore merged into Report Preparation to maintain clarity and reduce redundancy

c) Column Merged: CSS → web_design_and_development_tools

The original dataset contained a column named CSS. Since this information closely aligns with web design and development tools capabilities, it was merged into the web_design_and_development_tools column for consistency. This avoids duplication of concepts referring to web development skills.

d) Column Merged: Statistical Models → Bayesian

The dataset included a Statistical Models column. This content aligned closely with the purpose of the Bayesian variable. Information was therefore merged into Bayesian to maintain clarity and avoid redundancy in statistical methodology categorization.

e) Column Merged: Pattern Recognition → Clustering

The original dataset contained a column named Pattern Recognition. Since pattern recognition techniques include clustering methods, it was merged into the Clustering column for consistency and to consolidate machine learning concepts.

f) Column Merged: Experience in software lifecycle development → CS/SE Fundamentals

The dataset included an Experience in software lifecycle development column. This content aligned closely with the purpose of the CS/SE Fundamentals variable. Information was therefore merged to maintain clarity and better represent software engineering foundational knowledge.

g) Column Merged: Statistical analysis → Multivariate analysis

The original dataset contained a column named Statistical analysis. This content aligned closely with the purpose of the Multivariate analysis variable. Information was therefore merged into Multivariate analysis to consolidate statistical methodology concepts and reduce redundancy.

2. **Renamed columns**

The following variables were renamed to standardize naming conventions across all datasets

Original Name	Name in cleaned dataset
Excel	MS Excel
Looker	Looker studio
Familiar with GitHub	GitHub
Time series Forecasting	Time series Analysis
ML Models	Machine_Learning
Background in epidemiology	Epidemiology
OOP Concepts with applications	OOP(OBJECTED- ORIENTED PROGRAMMING)
Source	URL

3. Removal of Columns

Column Removed: Write Complex Queries

- The dataset contained a variable named Write Complex Queries.
- This information was removed because:
 - The dataset already includes other variables related to SQL and querying skills.

Columns Removed: Collaboration, Coding in any programming language, inventory management, FMCG sales data, Spot trends, Supply chain, Collecting data, Experience in Programming industrial level applications, Deep driven analyses, Market scan, Optum, Flatiron, Experience working with large, complex datasets, Attention_to_Detail, Independent_Project_Ownership, Actionable_Insights, Quality Assessment Metrics, Algorithms, Debugging Workflows, Experience applying RFC, A/B Test, and Model.

- The dataset contained multiple variables that were exclusive to original dataset and had no corresponding match in the other dataset.
- These variables were removed because:
 - They were not present in the merging dataset, making them incompatible for integration.
 - Including them would create inconsistency and missing values across the merged dataset.
 - They represented information unique to one dataset group only.

Group 3

- A new Excel sheet was generated after reorganizing the original dataset.
- Information related to the following variables did not appear in the new sheet: SAS, MATLAB, OLE/DB, Spreadsheets, Data_visualization, Data_warehouse, cloud_storage, Google_Cloud, AWS, Computer_vision, C, Linux/Unix, NoSQL, RDBMS, Oracle, SPL, web_design_and_development_tools, Google_Analytics, graphics_and_design_skills, Data_marketing, SEO, Content_Management, Alteryx, KNIME, Spotfire, S3, Redshift, Kafka, Storm, Bash, Hadoop, MPP_Platforms, Qlik, Pig, Hive, Map/Reduce, Impala, MongoDB, Elasticsearch, agile execution, Data_management, Data_mining, Data_science, Numerical_Analysis, Economic, Finance_Knowledge,

Investment_Knowledge, Debtor_reconciliation, Bayesian, Actuarial and risk analysis, Adaptability, AGE LIMIT, Airflow, Analytical_Skills, AngularJS, ANOVA, Apache Kafka, Apache_Beam, Apache, API, Argo, Automation, Azure, Biopharmaceutical development and manufacturing, biostatistics, Canva, Cassandra, CDISC ADaM data standard, CI/CD tools, Clinical research, Cloud Data Service, Clustering, CMC, Creating Dashboards, CRM systems (Salesforce/Hubspot/Zoho), CS/SE Fundamentals, D3, Dagster, Data Cleaning, Data Engineering Principles, Data Infrastructure, Data migration, Data validation, Data Security, Data structures, Data governance, Data_modelling, DataBricks, DAX, Design Data, DevOps, DigitalOcean, Docker, Documentation, DOE, EHR, Epidemiology, ERP, ETL/ELT, Experience scaling causal inference models, Exploratory Data Analysis (EDA), Familiar with dbt, Fast API, Fine-tuning, Flink, Forecasting, Foresee, Full-Stack Skills, GA4, GCP, GIS, Git, Github, GitLab, Global Exposure, Glue, Google Tag Manager, Healthcare Analytics, HTML, Hypothesis Testing, Innovalon, Interpretation skills, Jenkins, JMP, Julia, Jupyter notebooks, KNN, Kubeflow, Langchain, Languages, Large Scale data, Linear Regression, LLM, M Queries, MAR, MariaDB, Matplotlib, Medallion_Lakehouse_Architecture, Microsoft Fabric ecosystem, MicroStrategy, Mixed Models, MLflow, MLOps tools, MS Fabric, MS SQL Server, MS_Office, MS_Outlook, MS_Teams, Multivariate analysis, Navigate ambiguity, Network performance monitoring, NLP, NodeJS, ONE DRIVE, OOP (Object-Oriented Programming), OpinionLab, Organizational skills, OTHERS (LANGUAGES), Outlook, PL/SQL, Policy-level data interpretation, Portfolio management, Power Apps, Power Pivot, Power_Query, Predictive & Modeling Techniques, Preferred_Notice, Presto, Project management, Purview, PyTorch, Qualtrics, R Markdown, Random Forest, Redis, Report Preparation, Retrieval-Augmented Generation (RAG), Sampling methods, SAP, Scikit, SciPy, Scrum, Seaborn, Seldon, Semantic Kernel, SharePoint, Shell scripting, Snowflake, Solid grasp of applied statistics, Solr, Spark SQL, SPC Tools/Software, Springboot, SQL Server, SSIS, StakeholderManagementSkills, Stata, Stateless API, statistical tests, Streamlit, Tecton, Template Management, Teradata, Terraform, Time Series Analysis, TrinetX, T-SQL, Variance Component analysis, VBA, Warehouse Architecture, Weight & Biases, Windows, XGBoost, Zoom, Writing_Skills, Knowledge_in, Search_Term, Seniority Level, Mode, Specialization.

- In the original dataset, the minimum salary, maximum salary, and currency were all stored together in a single Salary column.

- Therefore, this was separated into three individual columns: Min_Salary, Max_Salary, and Currency.
- The new Excel sheet includes a Year column, which was generated by extracting the year component from the published date of each advertisement in the initial dataset.
- Job categories were assigned using the job titles.
- **Experience_Category** column: This column classifies experience requirements into four groups:
 - i. Unknown or Not Needed
 - ii. Two or Less Years
 - iii. More than 2 and Less than 5 Years
 - iv. More than 5 and Less than 10 Years
- Column removed: Source
 - Removed because the job posting link (URL) already provides this information (all advertisements were obtained through the internet).
- The Diploma column has been removed, and its information has been added to the Education_Qualification column in the format: Education (Diploma: ...) for consistency.
- The original dataset did not contain any entries for DateRetrieved; they were marked as NA.
- A Payment_Frequency column was generated by analyzing the salary column and identifying the frequency indicated within it.
- Renamed Columns

Original Excel sheet	New Excel sheet
Job Title	Job_title
Job Category	Job_Category
Source	URL

Date posted	DatePublished
Power BI	Microsoft Power BI
PySpark	pyspark
Sklearn	Scikit
Web Analytics tools	Web_Analytic_tools
Machine Learning	Machine_Learning
Looker Studio	Looker
Team work	team handling

Group 4

- A new Excel sheet was created by modifying the structure of the original Group 4 dataset.
- There were 106 variables in the original dataset of Group 4. They were Id, Job founder, Source of the advertisement, Job role, Company or Organization name, Experience, Location, Education requirement, Big Data, SAS, JMP, C++, R, SPSS, Apache, SSIS, Linux, Google Cloud, AngularJS, NodeJS, HTML, Java, Microsoft Fabric ecosystem, Python, No SQL, SQL, MS SQL, Spark SQL, T-SQL, Oracle, CI/CD tools, Azure SQL, Azure data service, SQL Server, Presto, AWS, Hadoop, AI, ML, Deep Learning, Hive, SPARK, Pyspark, Data migration, Data Cleaning, Data Infrastructure, Data transforming, SPC Tools/Software, Data Modeling, Power BI, Data Pipeline, Warehouse Architectures, Github, MySQL, ETL, Pytorch, Cloud Data Service, Azure Synapse Analytics, MS word, MS Excel, MS power point, Data visualization , Presentation skills, Analytical /Problem solving skills, Communication skills, Team work, English knowledge, Pandas, NumPy, Scikit learn, KNN, Time Series Analysis, Clustering, TensorFlow, Stateless API, Docker, Kubernetes, Airflow, Argo, Kubeflow, MLflow, Weight & Biases, Fast API, Scala, GCP, Adaptability, collaboration, Global Exposure, Reporting Skills, Share point , Out look, Teams, Canva Design, Template

Management, Content Creation , Linear Regression, ANOVA, DOE, Mixed models, Multivariate analysis, Variance component analysis, Bayesian methodology, Biopharmaceutical development and manufacturing, CMC, Clinical research, CDISC ADaM data standard, Database Management, Policy-level data interpretation, Data processing, Sampling methods, Applied statistics, statistical modelling, R Markdown, Jupyter notebooks, Data governance, kafka, HER, Healthcare Analytics, Epidemiology, Tableau, Qualtrics, OpinionLab, Foresee, and GIS.

- Then the similar variables but with different variable names were identified and renamed them according to the final variable names. They were as follows:

Original Variable name	Variable name in the cleaned dataset
Id	ID
Job founder	Consultant
Source of the advertisement	URL
Job role	Job title
Company or Organization name	Company
Experience	Experience category
Education Requirement	Education
Reporting skills	Report Preparation
Content creation	Content management
Bayesian methodology	Bayesian
Database management	RDBMS
Applied statistics	Solid grasp of Applied Statistics
Statistical modelling	Statistical analysis
Data transforming	Data management
Teams	Microsoft Teams

- Since there were many more columns in the final Excel sheet which were not in the initial Group 4 Excel sheet, they were added and filled with NAs.

- The columns BSc_needed, MSc_needed, and PhD_needed were not in the original Group 4 Excel sheet, those columns were added and filled using the Educational qualifications information.
- In the original Group 4 Excel sheet the Experience column was in a different format, that column was modified according to the final Excel sheet.
- In the original Group 4 Excel sheet under the variable Location, the city, country, and the mode were in -together. Those data were separated into three columns as City, Location (Country), and Mode according to the final Excel sheet.
- Since the variable Azure SQL, Azure data service, Azure synapse analytics, Collaboration, and Data processing were not in the final Excel sheet, the observations with those variables were added into the Knowledge_in column in the final Excel sheet.
- There were some missing values in the original Group 4 Excel sheet. They were filled with NAs.
- In the original Group 4 Excel sheet, the data in the Consultant variable (Job founder initially) were not in the same format. Therefore, that column was modified according to the format of the final Excel sheet.
- Many columns in the original Group 4 Excel sheet were not in the same alignments. Those things identified and corrected.

Group 5

- A new Excel sheet was created by including all the variables in Group 1 dataset and newly identified variables.
- There were 89 variables in the initial Group 5 dataset.
- The following columns in the initial dataset were renamed as below.

Variable name	Renamed as
Role(job name)	Job_title
MATLAB	MAtlab

Linux	Linux/Unix
PowerBI	Microsoft Power BI
Data governance	Data_governance
Agile	agile execution
MS 365 Suite	MS_Office
location(country)	Location
Model(online,offline)	Mode
year's experience	Experience category
STATA	Stata
CANVA	Canva
Source (weblink,newspaper..)	URL

- Then the data were filled into the new excel sheet.
- The new variables (columns) which were not in the initial dataset but were in the final dataset were filled as NA.
- A missing value on the column C# is reported as NA.
- Data in Experienced category were recoded as Two or less years, More than 2 and less than 5 years, More than 5 and less than 10 years, More than 10 years to match the format of data in other datasets.
- The data in Mode columns were Offline, Online, Online/ offline. They were recoded as below to match the format of data in other datasets.
 - Offline- Onsite
 - Online- Remote
 - Online/ offline- Hybrid
- There was a column named English_needed in the new format of the data set. That column was filled looking at the Languages column in the initial dataset. If English was mentioned as a language it was recoded as 1, otherwise 0.
- There were no columns named BSc_needed, special in Bsc, MSc_needed, PhD_needed in the initial dataset. In the new dataset they were filled looking at Educational_qualifications column. If the respective education qualification was mentioned in the job advertisement it was recoded as 1, otherwise 0.

Group 6

A new Excel sheet was generated after reorganizing the original dataset. The following changes and transformations were applied:

1. Original Dataset Overview

The original dataset contained 94 variables, which are as follows:

ID, Consultant, Source, URL, Search_Term, Date_Retrieved, Date_Published, Job_Title, Company, City, Province/State, Country, Career_Level, Employment_Type, Experience, Experience_Area, Salary, Currency, Payment_Freq, BSC_Needed, MSC_Needed, PhD_Needed, R, Python, Java, Scala, Unix/Linux, SPSS, Stata, SAS, JMP, DevOps, SQL, MySQL, MS_SQL, NoSQL, MongoDB, DB2, BigQuery, Looker, PLSQL / PLX, DataBricks, Seldon, MLflow, SpringBoot, Cloud, AWS, Azure, GCP, Snowflake, Kubeflow, Tecton, Jenkins, Kafka, Redis, Flink, Spark, Hadoop, Hive, Cassandra, Airflow, Apache_Beam, ETL/ELT, Data_Warehousing, Big_Data, Medallion_Lakehouse_Architecture, PowerBI, Tableau, MS_Excel, MS_Access, MS_Word, MS_PowerPoint, MS_Office, MS_Outlook, MS_Teams, Power_Query, Alteryx, Data_Visualization, AI, ML, NLP, Statistical_Knowledge, Analytical_Skill, Communication_Skills, Writing_Skills, Organizing_Skills, Problem_Solving_Skills, Teamwork, Leadership_Skills, Presentation_Skills, Collaboration_Skills, CS/SE_Fundamentals, Data_Mining

2. Additions and Replacements in the New Dataset

Columns that were not available in the original dataset were added in the new Excel sheet and marked as 'NA'. Examples of these new columns include:

Matlab, C#, OLE/DB, Spreadsheets, Deep Learning, Computer_vision, C++, C, RDBMS, Oracle, PHP, SPL, web_design_and_development_tools, Google_Analytics, graphics_and_design_skills, Data_marketing, SEO, Content_Management, KNIME, Spotfire, Redshift, Javascript, Data_Pipelines, MPP_Platforms, Qlik, Pig, Tensorflow, Map/Reduce, Impala, Elasticsearch, agile

execution, Data_management, pyspark, Data_science, Web_Analytic_tools, Numerical_Analysis, Economic, Finance_Knowledge, Investment_Knowledge, Debtor_reconciliation, Bayesian Optimization, Actuarial and risk analysis, Adaptability, AGE_LIMIT, AngularJS, ANOVA, Apache, API, Argo, Automation, Biopharmaceutical development and manufacturing, biostatistics, Canva, CDISC ADaM data standard, CI/CD tools, Clinical research, Cloud computing, Cloud Data Service, Clustering, CMC, Creating Dashboards, CRM systems (Salesforce/Hubspot/Zoho), D3, Dagster, Data Cleaning, Data Engineering Principles, Data Infrastructure, Data migration, Data Security, Data structures, Data validation, Data_governance, Data_modeling, English_needed, special in BSC, Job_Category, Max_Salary, Educational_qualifications, XGBoost, Zoom, Template Management, Teradata, Terraform, Time Series Analysis, TrinetX, T-SQL, Variance Component analysis, VBA, Warehouse Architecture, Weight & Biases, Windows, Stateless API, Statistical analysis, statistical tests, Streamlit, Synapse, Solid grasp of applied statistics, Solr, Spark SQL, SPC Tools/Software, , SSIS, Stakeholder Management Skills, Semantic Kernel, Sharepoint, Shell scripting, sklearn, Report Preparation, Retrieval-Augmented Generation (RAG), Sampling methods, SAP, Scikit, Scipy, Scrum, Seaborn, Predictive & Modeling Techniques, Preferred_Notice, Presto, Project Management, Purview, PyTorch, Qualtrics, R Markdown, Random Forest, Policy-level data interpretation, Portfolio management, Power Apps, Power Pivot, OTHERS(LANGUAGES), Outlook, Pandas, NodeJS, Numpy, ONE DRIVE, OOP (Object-Oriented Programming), OpinionLab, Multivariate analysis, Navigate ambiguity, Network performance monitoring, MLOps tools, MS Fabric, MS SQL Server, Metadata and master data management, Microsoft Fabric ecosystem, MicroStrategy, Mixed Models, LLM, M Queries, MAR, MariaDB, Matplotlib, Linear Regression, Logical Thinking, Kubernetes, Langchain, Languages, Large Scale data, Julia, Jupyter notebooks, Keras, GIS, Git, Github, GitLab, Global Exposure, Glue, Google Data Studio, Google Tag Manager, Healthcare Analytics, HTML, Hypothesis Testing, Innovalon, Interpretation skills, KNN, Forecasting, Foresee, Full-Stack Skills, GA4, Experience scaling causal inference models, Exploratory Data Analysis (EDA), Familiar with dbt, Fast API, Fine-tuning, DigitalOcean, Docker, Documentation, DOE, EHR, Employment Type (PART TIME/FULL TIME/internship/contract), Epidemiology, ERP, DAX, Decision Making, Design Data

3. Transformations Applied

3.1 Experience Column

The 'Experience' variable from the original dataset was transformed into an 'Experience_Category' column, classifying experience requirements into four groups:

1. Unknown or Not Needed
2. Two or Less Years
3. More than 2 and Less than 5 Years
4. More than 5 and Less than 10 Years

Additionally, data from the 'Experience_Area' column in the original dataset was included in the 'Knowledge' column in the new dataset.

3.2 Employment Category

The original 'Employment_Category' column included both employment mode and type. In the new dataset, the employment mode was separated into a 'Mode' column.

3.3 Qualification Requirements

Original dataset coding for qualifications:

- 0 = Not required
- 1 = Required
- 2 = Preferred

In the new dataset, '2' (Preferred) was replaced with '1' (Required).

3.4 Salary Column

The original dataset stored all salary information in a single column. In the new dataset, it was separated into:

- Min_Salary
- Max_Salary

3.5 Year Column

A new 'Year' column was created using the 'Date_Published' field from the original dataset. The corresponding year was extracted and recorded for each advertisement.

3.6 Handling Missing Columns

Any column not present in the original dataset was filled with 'NA' in the new Excel sheet.

This reorganization ensures a tidy, analysis-ready dataset with clear separation of variables, consistent coding, and additional useful metadata.

Group 7

- A new Excel sheet was created by modifying the structure of the initial dataset.
- There were twenty-three data variables in the initial data set such as ID, Consultant, Date Published, Job Category, Job Title, Mode, Company, City, Location, Educational Qualifications, R, Python, MS Word, MS Excel, Data Visualization, Machine Learning, SQL, Communication Skills, English Knowledge Skills, Salary, Experience, Source, URL.
- The “source” column was removed because the source was already identifiable through the URL column.
- The columns BSc_needed, MSc_needed, and PhD_needed were filled using the Educational Qualifications information from the original Excel sheet. Each column was marked with 1 if the qualification was required and 0 if it was not mentioned.
- In the original dataset, the minimum salary, maximum salary, and currency were all combined in a single salary column. These components were then split into three separate columns: Min_Salary, Max_Salary, and Currency.
- The year column in the new excel sheet was created using the published date from the initial dataset, with the corresponding year extracted and recorded for each advertisement.

Group 8

- A revised Excel sheet was developed by updating and reorganizing the original dataset. The modifications were carried out to correct errors, fill missing information, and improve the overall structure of the data.
- Information related to these variables did not appear.

MAtlab, OLE/DB, Data_visualization, Presentation_Skills, BigData, Data_warehouse, cloud_storage, Deep Learning, Computer_vision, NoSQL, RDBMS, MySQL, PHP, SPL,web_design_and_development_tools,Natural_Language_Processing(NLP),Google_Analyti cs, graphics_and_design_skills, Data_marketing, SEO, Content_Management, Alteryx, KNIME, Spotfire, S3, Redshift, Javascript, Storm, Bash, MPP_Platforms, Pig, Hive, Map/Reduce, Impala, MongoDB, Elasticsearch, agile execution, Data_science,Web_Analytic_tools, Numerical_Analysis, Economic, Investment_Knowledge, Debtor_reconciliation, Bayesian, AngularJS, ANOVA, Apache kafka, Apache_Beam, Aparche, API, Argo, Biopharmaceutical development and manufacturing, biostatistics, Canva, Cassandra, CDISC ADaM data standard, CI/CD tools, Clinical research, Cloud computing, Cloud Data Service, Clustering, CMC, CMC systems(salesforce/hubspot/zoho), D3, Dagster, Data Cleaning, Data migration, Data validation, Decision Making, Design Data, DevOps, DigitalOcean, Documentation, DOE, EHR, Epidemiology, ERP, Fast API, Flink, Foresee, Full-Stack Skills, GA4, GCP, GIS, Github, GitLab, Global Exposure, Glue, Google Data Studio, Google Tag Manager, HTML, Hypothesis Testing, Innovalon, Interpretation skills, Jenkins JMP, Julia, Jupyter notebooks, keras, KNN, Kubeflow, Languages, Large Scale data, Linear Regression, logical Thinking, MAR, MariaDB, Medallion LAkehouse_Architecture, Microsoft Fabric ecosystem, Mixed Models, MLflowt, MS Fabric, MS SQL Sever, MS_Office, MS_Outlook,Navigate ambiguity, NodeJS, ONE DRIVE, OpinionLab,Out look, PL/SQL, Power Apps, Power_Query, Preferred_Notice, Presto, Project management, Purview, Qualtrics, R Markdown, Redis, Sampling methods, SAP, Scripy, Scrum, Seldon, Share point, Shell scripting, sklearn, Snowflake, Solid grasp of applied statistics, Solr, Spark SQL, SPC Tools/Software, Springboot, SQL Server, SSIS, StakeholderManagementSkills, Stata, Stateless API, statistical tests, Synapse, Tecton, Template Management, Teradata,

Terraform, Time Series Analysis, TrinetX, T-SQL, Variance Component analysis, Warehouse Architecture, Weight & Biases, Writing_Skills, Knowledge_in ,Search_Term, Seniority Level

- In the original dataset, the minimum salary, maximum salary, and currency were all stored together in a single salary column.
- Therefore, we separated this into three individual columns: Min_Salary, Max_Salary, and Currency.
- The new Excel sheet includes a year column that was generated by extracting the year component from the published date of each advertisement in the initial dataset.
- Job categories were assigned using the job titles.
- The “minimum experience” variable in the original dataset was transformed into an “Experience_Category” column. This column classifies experience requirements into four groups:
 - i. Unknown or Not Needed
 - ii. Two or Less Years
 - iii. More than 2 and Less than 5 Years
 - iv. More than 5 and Less than 10 Years
- A payment frequency column was generated by analyzing the salary column and identifying the frequency indicated within it.
- The original dataset included several columns that were renamed to maintain consistency and achieve standardization throughout the dataset

Original Excel sheet	New Excel sheet
Work TYPE (ONSITE/REMOTE/HYBRID)	mode

Employment Type(PART TIME/FULL TME/internship)	Employment Type(PART TIME/FULL TME/internship/contract)
Educational qualification	Bsc , special in Bsc, MSC , PHD
others	OTHERS(LANGUAGES)
GOOGLE SHEETS	spreadsheets
TEAMS	MS_Teams
Scikit-learn (SKlearn)	Scikit
Power BI (BI tools)	PowerBI
Tableau (BI tools)	Tableau
Looker (data visualization tools)	Looker
dbt	Famililar with dbt
Advanced Multivariate & Exploratory Analysis	Multivariate analysis
Experimentation & Optimization	Optimization
Data handling	data managing
Core Statistical & Mathematical Analysis	Statistical analysis
Advanced Multivariate & Exploratory Analysis	multivariate analysis
GenAI (Generative AI	AI
Large Language Models (LLMs)	LLM
Nlp fundamenta	NLP
Data Processing & Automation	automation

Networking Protocols (TCP/IP, DNS, DHCP)	Data Infrastructure
Network Tools (Wireshark, Nagios, SolarWinds)	Network performance monitoring
Network Security & Compliance	data Security
Technology (Tech	CS/SE Fundamentals
Financial technology (Fintech)	finance
Demand planning	Data_mining
Portfolio management	potofolio management
Financial services and policy trends	Policy-level data interpretation
Work Ethic & Professionalism Responsibility	Leadership
Teamwork & Collaboration	team handling
Motivation & Growth Mindset Personal drive and attention	Adaptability

Group 9

- A new Excel sheet was generated after reorganizing the original dataset
- This dataset did not contain any entries for these variables, they were marked as NA.

DataRetrieved ,SAS, C#, MS Word, OLE/DB, Ms Access, Ms PowerPoint, Spreadsheets, Google_Cloud, Computer_vision,C++,C, NoSQL, RDBMS, MySQL, PHP, SPL, web_design_and_development_tools,Natural_Language_Processing(NLP),Google_Analytics, graphics_and_design_skills, Data_marketing,SEO, Content_Management, Alteryx,

KNIME, Spotfire, S3, Redshift, Javascript, Storm, Bash, Data_Pipelines, MPP_Platforms, Qlik, Pig, Tensorflow, Map/Reduce, Impala, MongoDB, Elasticsearch, agile execution, Data_management, pyspark, Data_mining, Data_science, Web_Analytic_tools, Numerical_Analysis, Economic, Finance_Knowledge, Investment_Knowledge, Debtor_reconciliation, Bayesian, Acturial and risk analysis, Adaptability, AGE LIMIT, Airflow, AngularJS, ANOVA, Apache, kafka, Apache, Beam, Aparche, API, Argo, Biopharmaceutical development and manufacturing, biostatistics, Canva, Cassandra, CDISC ADaM data standard, CI/CD tools, Clinical research, Cloud computing, Cloud Data Service, Clustering, CMC, Creating Dashboards, CS/SE Fundamentals, D3, Data Cleaning, Data Engineering Principles, Data Infrastructure, Data migration, data Security, Data structures, Data validation, Data_governance, Data_modeling, DataBriks, DAX, Decision Making, Design Data, DevOps, DigitalOcean, DOE, EHR, Epidemiology, ERP, Experience scaling casual inference models, Exploratory Data Analysis (EDA), Fast APIFine-tuning, Flink , Foresee, GA4, GCP, GIS, Git, Github , Global Exposure, Glue,Google Data Studio, Google Tag Manager, Healthcare Analytics, HTML, Hypothesis Testing, Innovalon, Interpretation skills, Jenkins JMP, Julia, Jupyter notebooks, keras, KNN, Kubeflow, Langchain , Languages, Large Scale data, Leadership, Linear Regression, logical Thinking, Looker, M Queries, MAR, MariaDB, Matplotlib, Medallion LAkehouse_Architecture, LLM, Meta data and master data management, Microsoft Fabric ecosystem, MicroStrategy, Mixed Models, MLflow, MLOps tools, MS Fabric, MS SQL Sever, MS_Office, MS_Outlook, MS_Teams, Multivariate analysis, Navigate ambiguity, Network performance monitoring, NLP, NodeJS, Numpy , ONE DRIVE, OOP(OBJECTED- ORIENTED PROGRAMMING), OpinionLab, Organizational skills, OTHERS(LANGUAGES), Out look, pandas, PL/SQL, Policy-level data interpretation, potofolio management, Power Apps, Power Pivot, Power_Query, Predictive & Modeling Techniques, Preferred_Notice, Presto, Project management, Purview, PyTorch, Qualtrics, R Markdown, Random Forest, Redis, Report Preparation, Retrieval-Augmented Generation (RAG), Sampling methods, SAP, Scikit, Scripy, Scrum, Seaborn, Seldon, Semantic Kernel, Share point, Shell scripting, sklearn, Snowflake, Solid grasp of applied statistics, Solr, Spark SQL, SPC Tools/Software, Springboot, SQL Server, SSIS, Stata, Stateless API, Statistical analysis, statistical tests, Streamlit, Synapse, Tecton, Template

Management, Teradata, Terraform, Time Series Analysis, TrinetX, T-SQL, Variance Component analysis, VBA, Warehouse Architecture, Weight & Biases, Windows, Writing_Skills, XGBoost, zoom, Knowledge_in , Educational_qualifications, Search_Term, special in Bsc, English_needed

- In the original dataset, the minimum salary, maximum salary, and currency were all stored together in a single salary column.
- Therefore, we separated this into three individual columns: Min_Salary, Max_Salary, and Currency.
- The new Excel sheet includes a year column that was generated by extracting the year component from the published date of each advertisement in the initial dataset.
- Job categories were assigned using the job titles.
- The “minimum experience” variable in the original dataset was transformed into an “Experience_Category” column. This column classifies experience requirements into four groups:
 - v. Unknown or Not Needed
 - vi. Two or Less Years
 - vii. More than 2 and Less than 5 Years
 - viii. More than 5 and Less than 10 Years
- A payment frequency column was generated by analyzing the salary column and identifying the frequency indicated within it.
- The original dataset included several columns that were renamed to maintain consistency and achieve standardization throughout the dataset

Original Excel sheet	New Excel sheet
Country	Location
WorkMode	Mode
Source	URL

Job type	Employment type
Analysis skills	Analytical Skills
Team work abilities	Team handling
Cloud analysis	Cloud storage
Excel	Ms Excel
Power BI/Tableau	Power BI
Power BI/Tableau	Tableau
SQL	MySQL
BigdataHadoop	Big Data
BigdataHadoop	Hadoop
Communication skills	Communication
Teamwork abilities	Team handling

Group 10

This document summarizes all cleaning steps applied to Dataset 10 before merging with the remaining nine datasets.

1. Removal of Columns

- Column removed: Remote
- The dataset contained both a Remote column and a Mode column.
- The “Remote” information was already captured within the Mode variable (e.g., Remote, Hybrid, On-site).
- To avoid duplication and contradictions, the Remote column was removed.

- Column removed: Source

- Removed because the job posting link (URL) already provides this information.

(All the advertisements were obtained through internet)

2. Renamed Columns

The following columns in the original dataset were renamed for consistency and standardization across all datasets:

Original Column Name	Name in the Combined Data Set
Power_BI	Microsoft Power BI
Excel	MS Excel
ETL	ETL/ELT Tools
CI_CD	CI/CD Tools
Scikit_lern	Scikit

3. Correction of the URL Column

Rows 15, 16, 17, 18, and 19 in the original dataset contained descriptions of the advertisement instead of actual job posting links.

Cleaning steps:

- Manually searched and identified correct URLs for rows 15, 16, 18, and 19.
- For row 17, the available information was insufficient to locate the original source. Therefore, the URL for this row remains missing.

4. Variable Alignment

The dataset was aligned with the standardized variable structure used across all 10 datasets. As part of this process, information for the following variables were extracted:

- Currency - filled based on the country of the job location.
- year - extracted from the DatePublished column.

5. Standardization of “Mode” Column Values

The original dataset contained inconsistent spellings and formats for the work mode.

Examples of inconsistencies:

- onsite
- Onsite
- On-Site
- on-site
- on site

All forms were standardized to:

On-site

6. Standardization of “Experience” Column Values

Some records contained experience descriptions without specifying the duration, such as:

- Experience in specific roles (e.g., Data Analyst)
- Experience in a small firm or fast-paced consultancy
- Previous experience mentioned without duration

Since this variable was intended to capture the number of years of experience, all non-quantified experience descriptions were replaced with NA.

7. Clarification and Interpretation of “Preferred Notice”

- The meaning of the “Preferred Notice” variable was unclear.
- We contacted the dataset owners directly for clarification.
- Definition confirmed: *It refers to the period the applicant is expected to serve as notice at their current job before joining the new position.* (i.e., the notice period an applicant must give to the current employer.)

8. Date Value Standardization

Date fields originally contained values in multiple formats, so they were converted to a consistent format (YYYY-MM-DD).

9. Handling missing values

All missing values across the dataset were replaced with NA to ensure consistency in handling incomplete data.

In the final stage, the final 10 datasets were manually combined in Excel via copy and paste. The following columns were removed from the final dataset because they duplicated information already present in other columns.

- OLE/DB, Computer_vision, SPL, MongoDB, Clustering, DAX, Design Data, DigitalOcean, DOE, HER, Fast API, Forecasting, GA4, GCP, Jenkins, KNN, Kubeflow, Langchain, Languages, logical Thinking, M Queries, MAR, MLflow, MS_Office, MS_Outlook, Navigate ambiguity, NodeJS, Out look, Power Pivot, Seaborn, Solid grasp of applied statistics, Spark SQL, Streamlit, Synapse.

The following columns were removed because they contained fewer than five ‘1’ values, where a ‘1’ indicates that the skill is mentioned in the job advertisement.

- Google_Analytics, SEO, KNIME, Spotfire, Storm, Pig, Impala, Elasticsearch, Debtor_reconciliation, AGE LIMIT, AngularJS, Argo, Biopharmaceutical development and manufacturing, Cassandra, CDISC ADaM data standard, CMC, CRM systems(salesforce/hubspot/zoho), D3, Dagster, Data validation, ERP, Experience scaling casual inference models, Flink, Foresee, GIS, Google Data Studio, Google Tag Manager, Innovalon, Linear Regression, MariaDB, Medallion_Lakehouse_Architecture, Mixed Models, ONE DRIVE, OpinionLab, OTHERS(LANGUAGES), Power Apps, Preferred_Notice, Presto, Purview, Qualtrics, Random Forest, Redis, Sampling methods, SAP, Scripy, Scrum, Seldon, Share point, Shell scripting, sklearn, Solr, SPC Tools/Software, Springboot, SQL Server, SSIS, Stateless API, statistical tests, Tecton, Template Management, Teradata, Terraform, TrinetX, Variance Component analysis, Warehouse Architecture, Weight & Biases

The column `graphics_and_design_skills` was removed from final dataset because they were not directly related or important to data science job roles.

The following columns were removed, and the data they contained was incorporated into other columns that captured the same or similar information.

- Apache kafka, Apache_Beam.

All the columns previously recorded as NA were changed to 0. The R code used for that is below. [DS.R](#)